

Data Structures Lab03 Description | AÖYS

```
class Lab03 {  
  
    public static void main(String[] args) {  
  
    }  
  
}  
  
interface MyList<E> {  
  
    int size();  
  
    boolean isEmpty();  
  
    void clear();  
  
    void addFirst(E e);  
  
    void addLast(E e);  
  
    void insertAt(int index, E e);  
  
    E removeFirst();  
  
    E removeLast();  
  
    E removeAt(int index);  
  
    E get(int index);  
  
    E set(int index, E e);  
  
    int indexOf(E o);  
  
    boolean remove(E o);  
  
}  
  
class Node<E> {  
  
    public E item;  
  
    public Node<E> next;  
  
    public Node(E item, Node<E> next) {
```

```
this.item = item;
```

```
this.next = next;
```

```
}
```

```
}
```

```
class SinglyLinkedList<E> implements MyList<E> {
```

```
    public Node<E> getHead() {
```

```
        return head;
```

```
    }
```

```
}
```

```
class CircularLinkedList<E> implements MyList<E> {
```

```
    public Node<E> getTail() {
```

```
        return tail;
```

```
    }
```

```
}
```

```
class DNode<E> {
```

```
    public E item;
```

```
    public DNode<E> prev;
```

```
    public DNode<E> next;
```

```
    public DNode(E item, DNode<E> prev, DNode<E> next) {
```

```
        this.item = item;
```

```
        this.prev = prev;
```

```
        this.next = next;
```

```
    }
```

```
}
```

```
class DoublyLinkedList<E> implements MyList<E> {
```

```
    public DNode<E> getHead() {
```

```
        return head;
```

```
    }
```

```
public DNode<E> getTail() {  
  
    return tail;  
  
}  
  
}
```